

days – longer than a journey to the moon and back. With few exceptions, the other systems in the spacecraft operated as they should.

The next flight, Apollo 8, would be the first time a manned mission went as far as the moon. Unofficially titled ‘Around the Moon and Back’, the spacecraft launched on December 21, 1968 carrying astronauts Frank Borman, James Lovell Jr. and William Anders. After reaching the Earth’s orbit, they further boosted towards the moon. The mission also proved the first mid-course correction of a manned spacecraft. Apollo 8 finally reached lunar orbit, where the crew conducted necessary tests and also broadcast a Christmas message to television sets around the world.

Apollo 9 launched on March 3, 1969, into a 190-kilometre Earth orbit with Commander James McDivitt, Command Module Pilot David Scott and Lunar Module Pilot Russell Schweickart aboard. The primary objective of the mission was an Earth-orbital engineering test of the first crewed Lunar Module. The flight plan’s top priority was the CSM and LM rendezvous and docking. This was performed twice – once while the LM was still attached to the Saturn rocket, and again when the LM was active.

Apollo 10 launched on May 18, 1969, with crew members Commander Thomas Stafford, Command Module Pilot John Young and Lunar Module Pilot Eugene Cernan. The launch trajectory had been so satisfactory that only one of four midcourse corrections was needed.

The Apollo 10 mission encompassed all aspects of an actual crewed lunar landing, except the landing. It was the first flight of a complete, crewed Apollo spacecraft to operate around the moon. Objectives included a scheduled eight-hour lunar orbit of the separated Lunar Module and descent to about 15 kilometres off the moon’s surface before ascending for rendezvous and docking with the Command and Service Module. Pertinent data to be gathered in this landing rehearsal dealt with the lunar potential, or gravitational effect, to refine the Earth-based crewed spaceflight network tracking techniques, and to check out LM programmed trajectories and radar, and lunar flight control systems. Twelve television transmissions to Earth were planned, including the first colour TV pictures to Earth of the moon’s surface. All mission objectives were achieved.

Apollo 11

With every aspect of the programmes tested, it was now time for NASA to embark on its most ambitious mission yet: an actual landing on the moon.

Apollo 11 launched from Cape Canaveral on July 16, 1969, carrying Com-